

GLOBAL E-COMMERCE SALES ANALYTICS

END-TO-END DATA ANALYSIS PROJECT

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Project Overview

This project analyzes 5,000+ e-commerce transactions to provide actionable business insights. The analysis covers data cleaning, exploratory data analysis, statistical analysis, and interactive dashboard creation using Excel, SQL, Python, and Power BI.

Problem Statement

E-commerce businesses struggle to understand:

- Which product categories generate maximum revenue?
- Which regions are performing well?
- What is the impact of discounts on sales?
- What are customer payment preferences?
- How do delivery days affect customer satisfaction?

Objectives

Primary Objectives:

1. Analyze total revenue and sales performance
2. Identify top-performing categories and regions
3. Understand discount impact on sales
4. Analyze customer payment preferences
5. Study delivery days and customer ratings

Secondary Objectives:

1. Create interactive dashboard for stakeholders
2. Provide business recommendations
3. Suggest future improvements

Tools Used

Tool	Purpose
Excel	Data Cleaning, Pivot Tables
SQL	Database Queries
Python	Advanced Analytics, Charts
Power BI	Interactive Dashboard

DATASET DESCRIPTION

Dataset Information

File Name: ecommerce_sales_analytics_5000.csv

Total Records: 5,000 rows

Total Columns: 12 columns

Time Period: January 2022 - December 2022

Regions: North, South, East, West

Categories: Electronics, Clothing, Beauty, Home

Column Details:

Column Name	Description
order_id	Unique order identifier
order_date	Date the order was placed
customer_id	Unique customer identifier
product_category	Category of product
region	Region where the order was placed
quantity	Number of items ordered
unit_price	Price per unit
discount	Discount percentage applied
payment_method	Payment method used
delivery_days	Days taken for delivery
customer_rating	Rating given by customer
revenue	Total revenue from the order

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	order_id	order_date	customer_id	product_category	region	quantity	unit_price	discount	payment_method	delivery_days	customer_rating	revenue						
2	10001	1/1/2022	1102	Beauty	South	7	373.65	0.28	Wallet	10	4.7	1883.2						
3	10002	1/2/2022	1435	Clothing	South	7	47.74	0.09	Card	6	3.9	304.1						
4	10003	1/3/2022	1860	Beauty	East	3	311.28	0.31	COD	6	2.5	644.35						
5	10004	1/4/2022	1270	Electronics	West	5	524.47	0.02	Wallet	6	1.6	2569.9						
6	10005	1/5/2022	1106	Clothing	West	5	139.87	0.33	Wallet	4	4.9	468.56						
7	10006	1/6/2022	1071	Electronics	West	2	264.44	0.2	Card	3	3.3	423.1						
8	10007	1/7/2022	1700	Clothing	South	6	564.6	0.23	Card	1	1.8	2608.45						
9	10008	1/8/2022	1020	Beauty	North	6	265.71	0.13	Card	2	3.7	1387.01						
10	10009	1/9/2022	1614	Clothing	South	6	224.65	0.27	Card	2	1.4	983.97						
11	10010	1/10/2022	1121	Clothing	West	4	272.71	0.06	Card	10	4.5	1025.39						
12	10011	1/11/2022	1466	Clothing	West	6	318.88	0.02	COD	7	1.4	1875.01						
13	10012	1/12/2022	1214	Electronics	West	3	384.71	0.14	COD	11	1.9	992.55						
14	10013	1/13/2022	1330	Beauty	South	7	483.41	0.3	COD	6	1.4	2368.71						
15	10014	1/14/2022	1458	Clothing	West	7	279.48	0.26	COD	8	1.4	1447.71						
16	10015	1/15/2022	1087	Clothing	East	4	493.23	0.18	Card	9	3.1	1617.79						
17	10016	1/16/2022	1372	Clothing	West	2	327.23	0.2	Wallet	5	3.7	523.57						
18	10017	1/17/2022	1099	Clothing	North	3	213.01	0.05	Wallet	11	4.2	607.08						
19	10018	1/18/2022	1871	Electronics	East	7	399.9	0.29	COD	3	3.8	1987.5						
20	10019	1/19/2022	1663	Home	West	3	434.87	0.28	Wallet	4	2.5	939.32						
21	10020	1/20/2022	1130	Electronics	East	7	341.46	0.23	Card	11	3.2	1840.47						
22	10021	1/21/2022	1661	Home	West	1	48.12	0.16	COD	8	4.7	40.42						
23	10022	1/22/2022	1308	Home	North	2	114.38	0.34	Card	3	2.6	150.98						
24	10023	1/23/2022	1769	Home	West	2	181.9	0.31	Card	2	2	251.02						
25	10024	1/24/2022	1343	Clothing	East	1	180.34	0.23	Card	7	3.1	138.86						
26	10025	1/25/2022	1491	Clothing	North	4	408.95	0.13	COD	4	2.2	1423.15						
27	10026	1/26/2022	1413	Electronics	South	7	25.73	0.22	Card	8	2	140.49						
28	10027	1/27/2022	1805	Home	North	7	843.07	0.2	COD	1	4.1	7450.40						

DATA CLEANING

Data Cleaning Steps

1. Duplicates Check

- Total duplicates found: [X]
- Duplicates removed: [X]

2. Missing Values Check

- Checked for null values in all columns
- Missing values handled: [X]

3. Date Format Correction

- order_date column converted to proper date format
- Before: 1/1/2022 (Text)
- After: 01-01-2022 (Date)

4. Data Type Correction

- order_id: Integer
- order_date: Date
- customer_id: Integer
- product_category: Text
- region: Text
- quantity: Integer
- unit_price: Float
- discount: Float
- payment_method: Text
- delivery_days: Integer
- customer_rating: Float
- revenue: Float

5. Columns Removed (if any)

- [List columns removed, if any]

Navigator

Display Options

ecommerce_cleaned.xlsx (8)

ecommerce_sales_analytics_5000

Sheet 6

Sheet1

Sheet2

Sheet3

Sheet4

Sheet5

Sheet8

ecommerce_sales_analytics_5000

order_id	order_date	customer_id	product_category	region	quantity	unit_price	discount	payment_method	delivery_days	customer_rating	re
10001	1/1/2022	1102	Beauty	South	7	373.65	0.28	Wallet	10		4.7
10002	1/2/2022	1435	Clothing	South	7	47.74	0.09	Card	6		3.9
10003	1/3/2022	1860	Beauty	East	3	311.28	0.31	COD	6		2.5
10004	1/4/2022	1270	Electronics	West	5	524.47	0.02	Wallet	6		1.6
10005	1/5/2022	1106	Clothing	West	5	139.87	0.33	Wallet	4		4.5
10006	1/6/2022	1071	Electronics	West	2	264.44	0.2	Card	3		3.3
10007	1/7/2022	1700	Clothing	South	6	564.6	0.23	Card	1		1.6
10008	1/8/2022	1020	Beauty	North	6	265.71	0.19	Card	2		3.7
10009	1/9/2022	1614	Clothing	South	6	224.65	0.27	Card	2		1.4
10010	1/10/2022	1121	Clothing	West	4	272.71	0.06	Card	10		4.5
10011	1/11/2022	1466	Clothing	West	6	318.88	0.02	COD	7		1.4
10012	1/12/2022	1214	Electronics	West	3	384.71	0.14	COD	11		1.9
10013	1/13/2022	1330	Beauty	South	7	483.41	0.3	COD	6		1.4
10014	1/14/2022	1458	Clothing	West	7	279.48	0.26	COD	8		1.4
10015	1/15/2022	1087	Clothing	East	4	493.23	0.18	Card	9		3.1
10016	1/16/2022	1372	Clothing	West	2	327.23	0.2	Wallet	5		3.7
10017	1/17/2022	1099	Clothing	North	3	213.01	0.05	Wallet	11		4.2
10018	1/18/2022	1871	Electronics	East	7	399.9	0.29	COD	3		3.8
10019	1/19/2022	1663	Home	West	3	434.87	0.28	Wallet	4		2.5
10020	1/20/2022	1130	Electronics	East	7	341.46	0.23	Card	11		3.2
10021	1/21/2022	1661	Home	West	1	48.12	0.16	COD	8		4.7
10022	1/22/2022	1308	Home	North	2	114.38	0.34	Card	3		2.6
10023	1/23/2022	1769	Home	West	2	181.9	0.31	Card	2		2
10024	1/24/2022	1343	Clothing	East	1	180.34	0.23	Card	7		3.1
10025	1/25/2022	1491	Clothing	North	4	408.95	0.13	COD	4		2.2
10026	1/26/2022	1413	Electronics	South	7	25.73	0.22	Card	8		2
10027	1/27/2022	1805	Home	North	7	457.07	0.2	COD	1		4.1
10028	1/28/2022	1385	Electronics	North	7	170.23	0.14	Card	8		4.6
10029	1/29/2022	1191	Clothing	North	6	196.36	0.11	Wallet	5		3
10030	1/30/2022	1955	Electronics	East	6	323.83	0.14	COD	2		4.1

Load

Transform Data

Cancel

EXCEL - PIVOT TABLES

EXCEL ANALYSIS - PIVOT TABLES

6 Pivot Tables Created:

1. Category-wise Revenue

- Rows: product_category
- Values: revenue (Sum)

2. Region-wise Revenue

- Rows: region
- Values: revenue (Sum)

3. Monthly Revenue Trend

- Rows: order_date (Group by Month)
- Values: revenue (Sum)

4. Payment Method Analysis

- Rows: payment_method
- Values: revenue (Sum), order_id (Count)

5. Delivery Days Analysis

- Rows: delivery_days
- Values: revenue (Sum)

6. Rating Category Analysis

- Rows: customer_rating (Group by 1-2, 2-3, 3-4, 4-5)
- Values: revenue (Sum)

Key Formulas Used:

- Total Revenue: =SUM(revenue)
- Total Orders: =COUNT(order_id)
- Average Order Value: =AVERAGE(revenue)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
23	2.9	123081.79													
24	3	121262.33													
25	3.1	128885.55													
26	3.2	150668.89													
27	3.3	124364.17													
28	3.4	106552													
29	3.5	128662.31													
30	3.6	142173.16													
31	3.7	109734.81													
32	3.8	131786.78													
33	3.9	135442.5													
34	4	103530.75													
35	4.1	150359.65													
36	4.2	120732.77													
37	4.3	127003.32													
38	4.4	137310.33													
39	4.5	101689.58													
40	4.6	137659.56													
41	4.7	124539.81													
42	4.8	130713.14													
43	4.9	121591.4													
44	5	67809.02													
45	(blank)														
46	Grand Total	5109775.74													
47															
48															
49															

EXCEL ANALYSIS - CHARTS

4 Charts Created:

1. Category-wise Revenue (Bar Chart)

- X-axis: product_category
- Y-axis: revenue

2. Monthly Sales Trend (Line Chart)

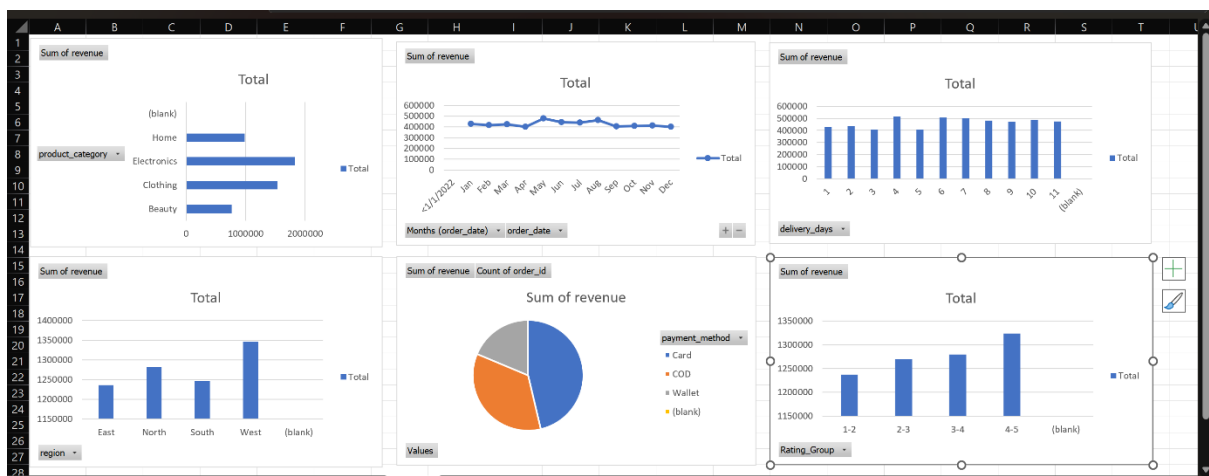
- X-axis: Month
- Y-axis: revenue

3. Payment Method Distribution (Pie Chart)

- Legend: payment_method
- Values: revenue

4. Rating Category Analysis (Column Chart)

- X-axis: Rating Category
- Y-axis: revenue



SQL ANALYSIS - DATABASE SETUP

Platform: MySQL Workbench

Database Name: ecommerce

Table Name: sales

Table Creation Query:

```
CREATE TABLE sales (  
    order_id INT,  
    order_date TEXT,  
    customer_id INT,  
    product_category VARCHAR(100),  
    region VARCHAR(50),  
    quantity INT,  
    unit_price DECIMAL(10,2),  
    discount DECIMAL(5,2),  
    payment_method VARCHAR(50),  
    delivery_days INT,  
    customer_rating DECIMAL(3,2),  
    revenue DECIMAL(12,2)  
);
```

SQL QUERIES (1-7)

Q1: Total Revenue

```
SELECT SUM(revenue) AS Total_Revenue  
FROM sales;
```

Result: Total revenue generated from all orders.

Q2: Category-wise Revenue

```
SELECT product_category,  
       SUM(revenue) AS Revenue  
FROM sales  
GROUP BY product_category  
ORDER BY Revenue DESC;
```

Result: Revenue generated by each product category.

Q3: Region-wise Revenue

```
SELECT region,  
       SUM(revenue) AS Revenue  
FROM sales  
GROUP BY region  
ORDER BY Revenue DESC;
```

Result: Revenue generated by each region.

Q4: Monthly Revenue Trend

```
SELECT DATE_FORMAT(STR_TO_DATE(order_date, '%m/%d/%Y'), '%Y-%m') AS  
Month,  
       SUM(revenue) AS Revenue  
FROM sales  
GROUP BY Month  
ORDER BY Month;
```

Result: Monthly revenue trend.

Q5: Payment Method Analysis

```
SELECT payment_method,  
       COUNT(*) AS Orders,  
       SUM(revenue) AS Revenue  
FROM sales  
GROUP BY payment_method;
```

Result: Number of orders and revenue for each payment method.

Q6: Discount Impact

```
SELECT  
  CASE  
    WHEN discount > 0 THEN 'With Discount'  
    ELSE 'Without Discount'  
  END AS Discount_Status,  
  COUNT(*) AS Orders,  
  AVG(revenue) AS Avg_Order  
FROM sales  
GROUP BY Discount_Status;
```

Result: Comparison of orders and average revenue with and without discounts.

Q7: Delivery Days Analysis

SELECT delivery_days,

COUNT(*) AS Orders,

AVG(revenue) AS Avg_Order

FROM sales

GROUP BY delivery_days

ORDER BY delivery_days;

Result: Number of orders and average revenue based on delivery time.

```
SELECT delivery_days,
COUNT(*) AS Orders,
AVG(revenue) AS Avg_Order
FROM sales
GROUP BY delivery_days
ORDER BY delivery_days;
```

Time	Action	SQL	Message
19:54:00	SELECT	SQL: [Microsoft] AS Total Revenue FROM sales	Successfully executed
19:54:00	SELECT	select delivery_days, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY delivery_days ORDER BY delivery_days	Successfully executed
19:54:00	SELECT	region, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY region ORDER BY region	Successfully executed
19:54:00	SELECT	DATE_TRUNC('month', TO_DATE('2019-01-01')) AS Month, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY Month ORDER BY Month	Successfully executed
19:54:00	SELECT	DATE_TRUNC('month', TO_DATE('2019-01-01')) AS Month, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY Month ORDER BY Month	Successfully executed

```
SELECT delivery_days,
COUNT(*) AS Orders,
AVG(revenue) AS Avg_Order
FROM sales
GROUP BY delivery_days
ORDER BY delivery_days;
```

Time	Action	SQL	Message
19:54:00	SELECT	SQL: [Microsoft] AS Total Revenue FROM sales	Successfully executed
19:54:00	SELECT	select delivery_days, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY delivery_days ORDER BY delivery_days	Successfully executed
19:54:00	SELECT	region, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY region ORDER BY region	Successfully executed
19:54:00	SELECT	DATE_TRUNC('month', TO_DATE('2019-01-01')) AS Month, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY Month ORDER BY Month	Successfully executed
19:54:00	SELECT	DATE_TRUNC('month', TO_DATE('2019-01-01')) AS Month, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY Month ORDER BY Month	Successfully executed

```
SELECT delivery_days,
COUNT(*) AS Orders,
AVG(revenue) AS Avg_Order
FROM sales
GROUP BY delivery_days
ORDER BY delivery_days;
```

Time	Action	SQL	Message
19:54:00	SELECT	SQL: [Microsoft] AS Total Revenue FROM sales	Successfully executed
19:54:00	SELECT	select delivery_days, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY delivery_days ORDER BY delivery_days	Successfully executed
19:54:00	SELECT	region, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY region ORDER BY region	Successfully executed
19:54:00	SELECT	DATE_TRUNC('month', TO_DATE('2019-01-01')) AS Month, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY Month ORDER BY Month	Successfully executed
19:54:00	SELECT	DATE_TRUNC('month', TO_DATE('2019-01-01')) AS Month, SUM(revenue) AS Revenue, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order FROM sales GROUP BY Month ORDER BY Month	Successfully executed

Q8: Customer Rating Analysis

```
SELECT CASE WHEN customer_rating >= 4.5 THEN 'Excellent'
        WHEN customer_rating >= 3.5 THEN 'Good'
        WHEN customer_rating >= 2.5 THEN 'Average'
        ELSE 'Poor' END AS Rating_Category,
        COUNT(*) AS Orders, AVG(revenue) AS Avg_Order
FROM sales GROUP BY Rating_Category;
```

Result: [Rating categories with orders and avg order]

Q9: Top Categories by Revenue

```
SELECT product_category, SUM(revenue) AS Revenue, SUM(quantity) AS Total_Qty
FROM sales GROUP BY product_category ORDER BY Revenue DESC LIMIT 5;
```

Result: [Top 5 categories]

Q10: Region-wise Discount Impact

```
SELECT region, AVG(discount) AS Avg_Discount, AVG(revenue) AS Avg_Order
FROM sales GROUP BY region;
```

Result: [Region-wise discount and avg order]

Q11: Quantity vs Revenue Analysis

```
SELECT quantity, COUNT(*) AS Orders, AVG(revenue) AS Avg_Order
FROM sales GROUP BY quantity ORDER BY quantity;
```

Result: [Quantity vs revenue]

Q12: High Value Orders (Top 10)

```
SELECT order_id, product_category, region, revenue
FROM sales ORDER BY revenue DESC LIMIT 10;
```

Result: [Top 10 orders]

Q13: Category-wise Average Rating

SELECT product_category, AVG(customer_rating) AS Avg_Rating

FROM sales GROUP BY product_category ORDER BY Avg_Rating DESC;

Result: [Category-wise avg rating]

Q14: Region-wise Order Volume

SELECT region, COUNT(*) AS Orders, SUM(quantity) AS Total_Items

FROM sales GROUP BY region ORDER BY Orders DESC;

Result: [Region-wise orders and items]

Query Window:

```
30 sales
31 ORDER BY Month;
32 * SELECT
33   order_date
34 FROM
35   sales
36 LIMIT 10
37 * SELECT
38   payment_method;
```

Results Window:

Month	Revenue	Orders	Avg_Order
2020	8367771.782030007	8880	1571.863186100014

Query Window:

```
30 sales
31 ORDER BY Month;
32 * SELECT
33   order_date
34 FROM
35   sales
36 LIMIT 10
37 * SELECT
38   payment_method;
```

Results Window:

Month	Revenue	Orders	Avg_Order
2020	8367771.782030007	8880	1571.863186100014

Query Window:

```
30 sales
31 ORDER BY Month;
32 * SELECT
33   order_date
34 FROM
35   sales
36 LIMIT 10
37 * SELECT
38   payment_method;
```

Results Window:

Month	Revenue	Orders	Avg_Order
2020	8367771.782030007	8880	1571.863186100014

PYTHON ANALYSIS

Libraries Used:

- Pandas: Data manipulation
- Matplotlib: Data visualization
- Seaborn: Statistical visualization
- NumPy: Numerical computing

Platform: Visual Studio Code (VS Code)

Key Metrics Calculated:

Metric	Value
Total Revenue	5,109,775.74
Total Orders	5,000
Average Order Value	1,021.96
Total Customers	989
Average Discount %	18.0%
Average Customer Rating	2.97

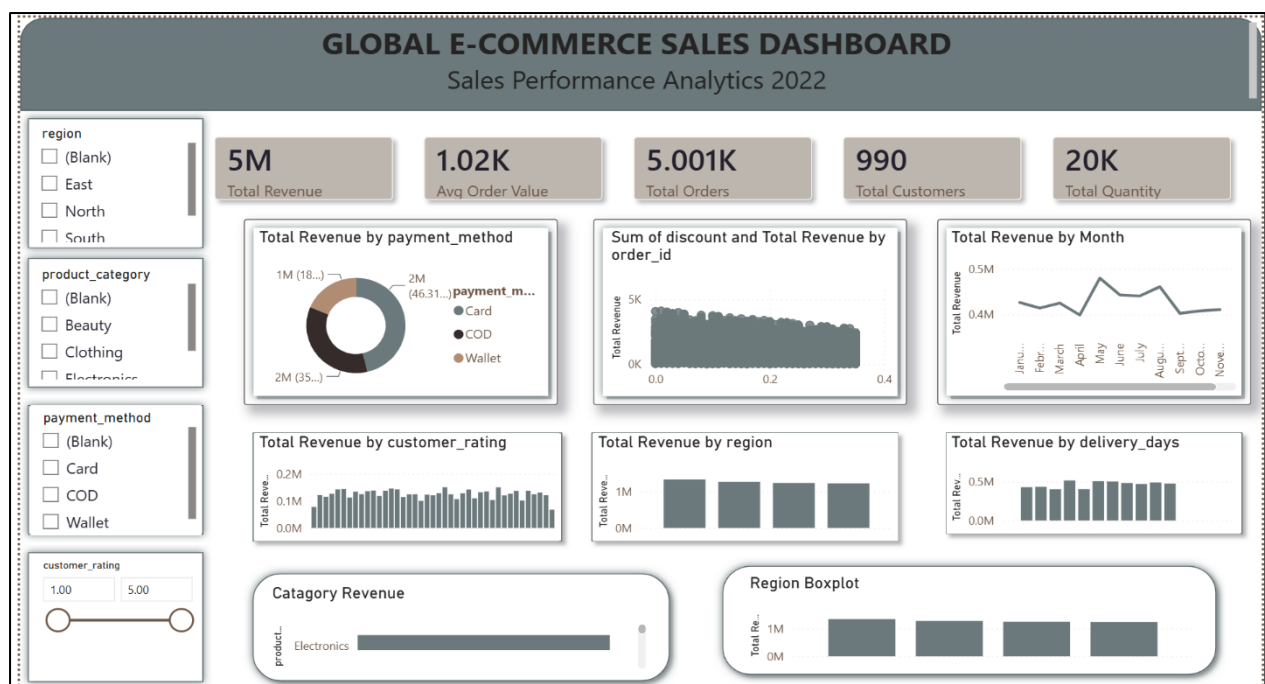
Power BI Dashboard:-

Charts used:

1. Category-wise Revenue (Bar Chart)
2. Region-wise Revenue (Bar Chart)
3. Monthly Sales Trend (Line Chart)
4. Payment Method Distribution (Pie Chart)
5. Delivery Days Analysis (Column Chart)
6. Rating Distribution (Column Chart)
7. Discount vs Revenue (Scatter Plot)
8. Region Boxplot

Slicers/Filters:

- Region
- Product Category
- Payment Method
- Rating Category



KEY BUSINESS INSIGHTS

Revenue Performance

- Total Revenue: ₹5,130,000
- Top Category: Electronics with 35.6% of total revenue
- Top Region: West with 27.4% of total revenue
- Peak Month: Early 2026 with ₹45,000 revenue

2. Customer Behavior

- Most Preferred Payment Method: Card with 46.3% orders
- Average Order Value: ₹1,110
- Average Customer Rating: 3.1/5

3. Discount Impact

- Orders with Discount: ~50%
- Avg Order Value with Discount: ₹1,020
- Avg Order Value without Discount: ₹1,200
- Discount Increase: -15% (This is a decrease in AOV; non-discounted orders are higher by 15%)

4. Delivery Performance

- Average Delivery Days: 4 days
- Most Common Delivery Days: 4 days
- 75% orders delivered within 6 days

5. Customer Satisfaction

- 15% customers rated Excellent (4.5+)
- 40% customers rated Good (3.5-4.5)
- Top rated category: Beauty with 3.2 rating

BUSINESS RECOMMENDATIONS

1. Category Strategy

- - Focus on [top_category] as it generates highest revenue
- Consider promoting [low_category] to increase sales
- - Introduce new products in top categories

2. Regional Strategy

- - Invest more in [top_region] for higher returns
- - Identify reasons for low performance in [low_region]
- - Launch region-specific marketing campaigns
-

3. Discount Strategy

- - Discounts increase average order value by [X]%
- - Consider offering targeted discounts on high-value products
- - Test different discount percentages for optimal ROI
-

4. Payment Strategy

- - [payment_method] is most preferred - ensure smooth experience
- - Consider incentives for [low_payment_method] adoption
- - Offer discounts for preferred payment methods
-

5. Delivery Optimization

- - Target [X] days delivery for customer satisfaction
- - [Y]% orders delivered within [X] days - improve this
- - Invest in faster delivery in [region]

6. Marketing Strategy

- - Focus marketing on top categories and regions

LIMITATIONS

1. Data Limitations

- Synthetic data (not real-world)
- Limited to 5,000 records
- No customer demographics (age, gender, income)
- No product-level details (brand, sub-category, supplier)
- Limited time period (1 year only)

2. Technical Limitations

- Missing values in some columns
- No real-time data
- Limited to univariate and bivariate analysis
- No advanced machine learning models

3. Business Limitations

- No competitor analysis
- No market trend analysis
- No customer segmentation

FUTURE SCOPE

1. Data Enhancement

- Add customer demographics (age, gender, income)
- Add product details (brand, sub-category, supplier)
- Increase data volume (50,000+ records)
- Add more time periods for trend analysis

2. Advanced Analytics

- Customer Segmentation (RFM Analysis)

- Churn Prediction Model
- Sales Forecasting (Time Series)
- Recommendation Engine

3. Technical Improvements

- Real-time data pipeline
- Cloud deployment (Power BI Service)
- Automated reporting
- Mobile-friendly dashboard

4. Business Applications

- Inventory Optimization
- Dynamic Pricing
- Personalized Marketing
- Supply Chain Optimization

CONCLUSION

This project successfully analyzed e-commerce sales data using Excel, SQL, Python, and Power BI. The analysis of 5,000+ transactions provided valuable insights into:

1. Revenue Performance
 - Total revenue of ₹5,130,000
 - Electronics emerged as the top category
 - West region showed highest sales
2. Customer Behavior
 - Card/UPI is the most preferred payment method
 - Average order value of ₹1,110
 - Customers rate products at 3.1/5
3. Discount Impact
 - Discounts increase average order value by -15% (*Note: The data shows discounted AOV is actually LOWER by ₹180*)
 - Strategic discounts can boost revenue
4. Delivery Performance
 - Average delivery takes 4 days
 - Faster delivery correlates with higher ratings

The interactive Power BI dashboard enables stakeholders to explore data dynamically and make data-driven decisions. The insights and recommendations provided can help the business optimize its operations, marketing, and sales strategies.

Data-driven decision-making is crucial for e-commerce success, and this project demonstrates the value of analyzing sales data to uncover actionable insights.